

Probability HW2

Problem #1

S = tomorrow is sunny, R = tomorrow is rainy
 F_1 = station 1 predicts sunny, F_2 = station 2 predicts rainy

$$P(F_1|S) = 0.9, P(F_1|R) = 0.1 \quad P(F_2|S) = 0.2 \quad P(F_2|R) = 0.8$$

$$P(F_1, F_2|S) = 0.18 \quad P(F_1, F_2|R) = 0.08$$

$$P(S|F_1, F_2) = \frac{P(S|F_1, F_2)}{P(F_1, F_2)} = \frac{P(F_1, F_2|S)P(S)}{P(F_1, F_2|S)P(S) + P(F_1, F_2|R)P(R)} = \frac{0.18 \cdot 0.5}{0.18 \cdot 0.5 + 0.08 \cdot 0.5} \approx \boxed{0.692}$$

Problem #2

Let A be surgery at high-volume, $\bar{A}$ be low-volume

Let B be died after two weeks of surgery

$$a) P(A) \cdot P(\bar{B}) = 0.77 \cdot 0.993 = 0.765 \quad \boxed{76.5\%}$$

$$b) \text{survived} = 1 - P(B) \quad P(B) = 0.77 \cdot 0.007 + 0.23 \cdot 0.031 = 0.01252$$

$$= 0.98748 \quad \boxed{98.748\%}$$

$$c) \text{one died and one survived} = (0.98748 \cdot 0.01252) \cdot 2 = 0.0247$$

$$= \boxed{2.47\%}$$

$$d) P(\text{survived} \geq 2) = (0.98748)^3 + (0.98748)^2(0.01252) \cdot 3$$

$$= 0.9629 + 0.0366 = 0.9995$$

$$= \boxed{99.95\%}$$

Problem #3

E = there is actual error, $\bar{E}$ = IRS shows error

$$P(E|F) = \frac{P(F|E)P(E)}{P(F|E)P(E) + P(F|\bar{E})P(\bar{E})}$$

$$P(F|E) = 0.7 \quad P(\bar{E}) = 0.2$$

$$P(F|\bar{E}) = 0.1$$

$$= \frac{0.7 \cdot 0.2}{0.7 \cdot 0.2 + 0.1 \cdot 0.8} = \frac{0.14}{0.22} = \boxed{0.6364}$$

Problem #4

a) Independent means $P(A \cap B) = P(A) \cdot P(B)$

$$P(A \cap \bar{B}) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(\bar{B})$$

$$b) P(E_1 \cup E_2 \dots \cup E_n) = 1 - P(\bar{E}_1 \cap \bar{E}_2 \dots \cap \bar{E}_n)$$

Since $E_1, \dots, E_n$ are all independent $\Rightarrow \bar{E}_1, \dots, \bar{E}_n$ are all independent

$$P(\bar{E}_1 \cap \bar{E}_2 \dots \bar{E}_n) = \prod_{i=1}^n (1 - P(E_i))$$

$$\therefore P(E_1 \cup E_2 \dots \cup E_n) = 1 - \prod_{i=1}^n (1 - P(E_i))$$

Problem #5

flashlights: 20% type 1, 30% type 2, 50% type 3

$F_1 = \text{type 1}, F_2 = \text{type 2}, F_3 = \text{type 3}$

go over 100 hrs: 0.7, 0.4, 0.3

$L = \text{lasted over 100 hrs}$

$$a) P(L) = 0.2 \cdot 0.7 + 0.3 \cdot 0.4 + 0.5 \cdot 0.3 = 0.41$$

$$b) P(F_1 | L) = \frac{P(L | F_1) P(F_1)}{P(L)} = \frac{0.7 \cdot 0.2}{0.41} = 0.3415$$

$$P(F_2 | L) = \frac{P(L | F_2) P(F_2)}{P(L)} = \frac{0.3 \cdot 0.4}{0.41} = 0.2927$$

$$P(F_3 | L) = \frac{P(L | F_3) P(F_3)}{P(L)} = \frac{0.5 \cdot 0.3}{0.41} = 0.3659$$

Problem #6

$E = \text{sum of dice equals 7}$

a) Let F be first dice lands on 4, E and F are independent if:

$$P(E \cap F) = P(E)P(F)$$

$$P(E \cap F) = \frac{1}{36} \quad P(E) = \frac{1}{6} \quad \left\{ \begin{array}{l} (2,5) \cdot 2 \\ (3,4) \cdot 2 \\ (4,3) \cdot 2 \\ (5,2) \cdot 2 \end{array} \right\} \quad P(F) = \frac{1}{6}$$

$$P(E \cap F) = P(E)P(F) \quad \therefore E \text{ is independent of } F$$

b) Let G be second dice lands on 3, E and G are independent if:

$$P(E \cap G) = P(E)P(G)$$

$$P(E \cap G) = \frac{1}{36} \quad P(E) = \frac{1}{6} \quad P(G) = \frac{1}{6}$$

$$P(E \cap G) = P(E)P(G) \quad \therefore E \text{ is independent of } G$$

Problem #7

$$a) P(A \cup B) = P(A) + P(B) = 0.5$$

$$b) P(A \cup B) = P(A) + P(B) - P(A)P(B) = 0.44$$

$$c) P(A \cap B \cap C) = P(A)P(B)P(C) = 0.024$$

$$d) P(A \cap B \cap C) = 0$$